Lecture 4 2025-09-09

Taylor's theorem $f \in C^n([a,b])$

$$f(x) = P_n(x) + R_n(x)$$

$$P_n(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2 + \dots + \frac{f^{(n)}(x_0)(x-x_0)^n}{n!}$$

$$R_n(x) = \frac{f^{(n+1)}(\xi(x))}{(n+1)!}(x-x_0)^{n+1} \qquad \xi(x) \text{ between } x \text{ and } x_0$$

Example 3 from book: $f(x) = \cos x$ & $x_0 = 0$

We want to find P_2 and estimate R_2 (P_3 & R_3)

$$f'(x) = -\sin x \quad f''(x) = -\cos x \quad f'''(x) = \sin x \quad f^{(4)}(x) = \cos x$$

$$P_2(x) = f(0) + f'(0)x + \frac{1}{2} f''(0)x^2$$
$$= 1 + 0 + \left(-\frac{1}{2}\right)x^2 = 1 - \frac{x^2}{2}$$

$$R_2(x) = \frac{1}{3!} f'''(\xi(x))\, x^3$$

$$P_3(x) = f(0) + f'(0)x + \frac{1}{2} f''(0)x^2 + \frac{1}{6} f'''(0)x^3$$
$$= 1 - \frac{1}{2}x^2 + 0 = P_2(x)$$

$$R_3(x) = \frac{1}{4!} f^{(4)}(\xi(x))\, x^4 = R_2(x)$$

We get a stronger estimate using the expression for R_3 instead of the one for R_2

e.g. $[0, \frac{1}{10}]$: $|R_2(x)| = \frac{1}{3!}|\sin(\zeta(x))|\,x^3 \le \frac{x^3}{6} \le \frac{(0.1)^3}{6}$

$= 0.0001667$

$|R_3(x)| = \frac{1}{4!}|\cos(\zeta(x))|x^4 \le \frac{(0.1)^4}{24} = 4.167 \cdot 10^{-6}$

stricter error bound

- When bounding integrals, it's better to bound R_n as a function of x (rather than the maximum error)

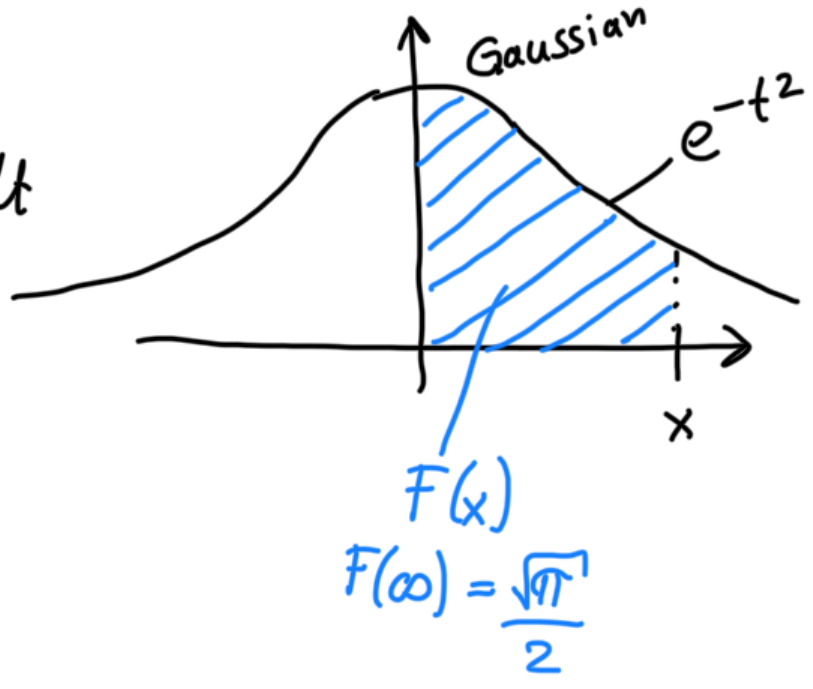
E.g. Estimate $|R_n(x)| \le Cx^{n+1}$ over $[0,b]$ yields a bound for $\int_0^b |R_n(x)|\,dx$ that is $n+2$ times smaller than using $|R_n(x)| \le Cb^{n+1}$

$$\int_0^b Cx^{n+1}\,dx = \frac{Cx^{n+2}}{n+2}\Big|_0^b = \frac{Cb^{n+2}}{n+2}$$

$$\int_0^b Cb^{n+1}\,dx = Cb^{n+2}$$

$(n+2)$ times larger

- Example: Let $F(x) = \int_0^x e^{-t^2}\,dt$



Goal:

Use $P_5(x)$ for $f(x) = e^{-x^2}$ to estimate $F(0.2)$

Solution: First find $P_5(x)$ and $R_5(x)$ with $x_0 = 0$

Trick: $e^y = 1 + y + \frac{y^2}{2} + \frac{y^3}{6} + \dots$

$\downarrow y = -x^2$

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6} + \dots$$

$$\underbrace{\qquad\qquad}_{=P_5(x)} \qquad \text{(and also } P_4(x)\text{)}$$

$$\tilde{f}(y) = e^y = 1 + y + \frac{y^2}{2} + \tilde{R}_2(y) \qquad \tilde{P}_2(y) = 1 + y + \frac{y^2}{2}$$

$$\tilde{R}_2(y) = \frac{e^{\xi(y)}}{3!} y^3$$

$$f(x) = e^{-x^2} = \tilde{f}(-x^2) = \underbrace{1 - x^2 + \frac{x^4}{2}}_{P_5(x)} \underbrace{- \frac{1}{6} e^{\xi(-x^2)} x^6}_{R_5(x)}$$

Remember: $R_5(x) = f(x) - P_5(x)$

Next: Estimate R_5 (What is $e^{\xi(-x^2)}$???)

We know e^y is increasing and $\xi(y)$ is between y and 0

So $\xi(-x^2)$ is between $-x^2$ and 0

$$e^{-x^2} \le e^{\xi(-x^2)} \le e^0 = 1 \quad \text{since } e^y \text{ is increasing}$$

Then we have $|R_5(x)| = \frac{1}{6} |e^{\xi(-x^2)}| \, x^6 \le \frac{x^6}{6}$

$$\int_0^{0.2} f(x)\,dx = \int_0^{0.2} P_5(x)\,dx + \int_0^{0.2} R_5(x)\,dx$$

error: $|\varepsilon| = \left| \int_0^{0.2} f(x)\,dx - \int_0^{0.2} P_5(x)\,dx \right| \le \left| \int_0^{0.2} R_5(x)\,dx \right|$

$$\le \int_0^{0.2} |R_5(x)|\,dx$$

$$\le \int_0^{0.2} \frac{x^6}{6}\,dx = \frac{x^7}{42}\Big|_0^{0.2} \approx 3.05 \cdot 10^{-7}$$

1.2. Round-off Errors and Computer Arithmetic

There are two major sources of error

I. Discretization Error

e.g. $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$ + error

$$\frac{du(x)}{dx} = \frac{u(x+h) - u(x-h)}{2h} + \text{error}$$

This type of error comes from approximating the original problem by a discrete or truncated problem.

II. Roundoff error

- Computers have finite precision.
  - No way to represent π or e as a binary or decimal expansion.
  - Multiply two 32-digit numbers will result in a 64 digit number ⇒ have to round to fit in 32 digits again.

Computers represent numbers in binary floating point arithmetic.

Decimal: $4.27 * 10^3$ means $(4 \cdot 10^0 + 2 * 10^{-1} + 7 * 10^{-2}) * 10^3$

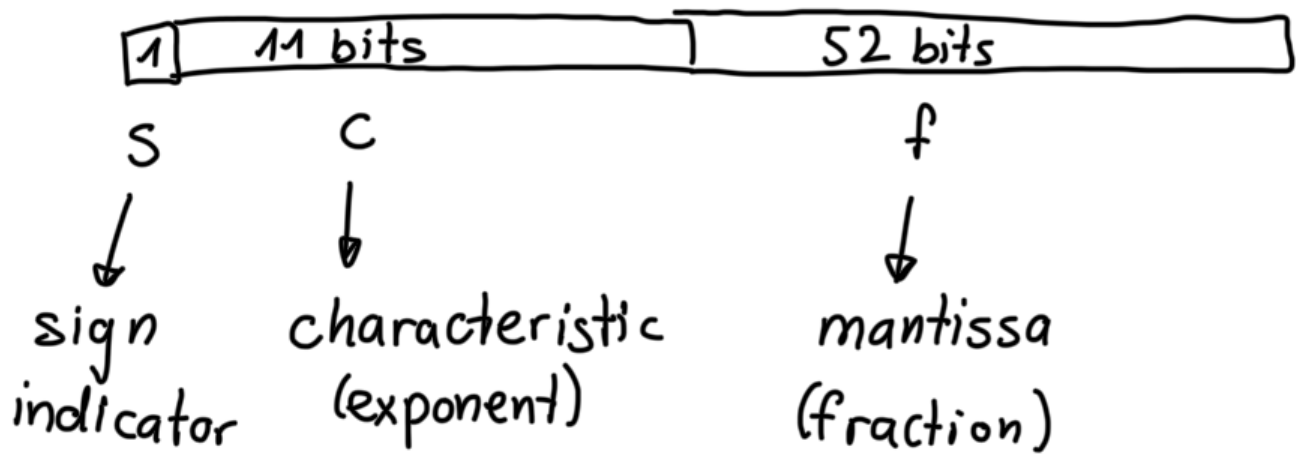
Binary: -1.0110×2^5 means $-(1 * 2^0 + 0 * 2^{-1} + 1 * 2^{-2} + 1 * 2^{-3}) * 2^5$

IEEE double precision standard (64 bits)

↳ Institute for Electrical and Electronic Engineers

- widely adopted standard

- default data type in Matlab, "double" in C
- 64 bits to represent a real number



represent number

$$x = (-1)^s \, 2^{c-1023} (1+f) \qquad \text{for} \quad 1 \le c \le 2046$$

base 2

$2^{11} - 1 = 2047$

$s = 0$: positive numbers

$s = -1$: negative numbers

general format $-1.0110 * 2^5$ → $\max(c-c_0, 1-c_0)$

double precision $c_0 = 1023$

$(-1)^s$ sign bit

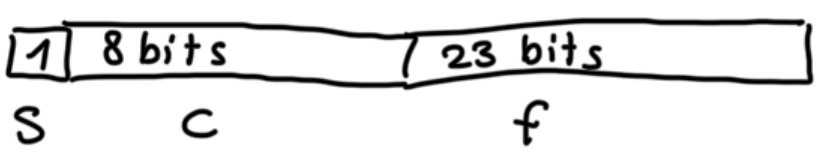
f

not stored assumed to be = 1 unless $c = 0$ when it is zero

"Special numbers"

$$x = \begin{cases} (-1)^s \, 2^{-1022} \, f, & c = 0 \\ \pm\infty & c = 2047, \; f = 0 \\ \text{NaN} & c = 2047, \; f \neq 0 \end{cases}$$

IEEE Single Precision: 32 bits ('float' in C)



$$x = \begin{cases} (-1)^s \, 2^{c-127} (1+f) & 1 \le c \le 254 \quad (c_0 = 127) \\ (-1)^s \, 2^{-126} \, f & c = 0 \end{cases}$$

$\pm\infty$, NaN

c= 255